Topics 6.6 – 6.9: MCQ Practice

$$4\ NH_3(g)\ +\ 5\ O_2(g)\ \rightarrow\ 4\ NO(g)\ +\ 6\ H_2O(l) \quad \Delta H^\circ = ?$$

1. It is observed that 13.0 kJ of energy is released when 1.00 g $NH_3(g)$ reacts completely with excess $O_2(g)$ under standard conditions according to the equation above. The value for the standard enthalpy change, ΔH°, for the reaction represented by the equation above is closest to

   (A) –884 kJ/mol$_{rxn}$

   (B) –221 kJ/mol$_{rxn}$

   (C) +221 kJ/mol$_{rxn}$

   (D) +884 kJ/mol$_{rxn}$

$$2\ C_2H_2(g)\ +\ 5\ O_2\ (g)\ \rightarrow\ 4\ CO_2(g)\ +\ 2\ H_2O(l) \quad \Delta H^\circ = -2600\ kJ/mol_{rxn}$$

2. A mixture of 3.0 mol $C_2H_2(g)$ and 6.0 mol $O_2(g)$ are added to a previously evacuated reaction vessel. The mixture is sparked, initiating a chemical reaction according to the equation above. Which of the following identifies the limiting reactant and the amount of heat (q) released under standard conditions?

	Limiting Reactant	q (kJ)
(A)	$C_2H_2(g)$	2600
(B)	$C_2H_2(g)$	3900
(C)	$O_2(g)$	2600
(D)	$O_2(g)$	3100

$$2\ Na(s)\ +\ 2\ H_2O(l)\ \rightarrow\ 2\ NaOH(aq)\ +\ H_2(g) \quad \Delta H^\circ = -368\ kJ/mol_{rxn}$$

3. When 2.30 g of Na(s) is added to 200. mL of water at 25°C in a calorimeter, all of the Na(s) reacts with water as represented by the equation above.

   Which of the following is true when the 2.30 g of Na(s) has completely reacted?

   (A) 18.4 kJ of energy is absorbed by the reaction.

   (B) 36.8. kJ of energy is absorbed by the reaction.

   (C) 18.4 kJ of energy is released by the reaction.

   (D) 36.8 kJ of energy is released by the reaction.

$$HNO_3(aq) + KOH(aq) \rightarrow H_2O(l) + KNO_3(aq) \quad \Delta H° = ?$$

4. In an experiment a student mixes a 50.0 mL sample of 1.0 *M* $HNO_3(aq)$ with a 50.0 mL sample of 1.0 *M* $KOH(aq)$ at 20.0°C in a coffee-cup calorimeter. Which of the following is the enthalpy change of the reaction represented above if the final temperature of the mixture is 27.0°C? Assume that the total mass of the mixture is 100. g and that the specific heat capacity of the mixture is 4.0 J/(g·°C).)

   (A) –56 kJ/mol$_{rxn}$

   (B) –2.8 kJ/mol$_{rxn}$

   (C) 2.8 kJ/mol$_{rxn}$

   (D) 56 kJ/mol$_{rxn}$

Substance	Chemical Equation for the Combustion Reaction
Butane, $C_4H_{10}(g)$	$C_4H_{10}(g) + \frac{13}{2} O_2(g) \rightarrow 4\ CO_2(g) + 5\ H_2O(l) \quad \Delta H° = -2880$ kJ/mol$_{rxn}$
Ethanol, $C_2H_5OH(l)$	$C_2H_5OH(l) + 3\ O_2(g) \rightarrow 2\ CO_2(g) + 3\ H_2O(l) \quad \Delta H° = -1370$ kJ/mol$_{rxn}$

5. A student sets up two experiments in which samples of $C_4H_{10}(g)$ and $C_2H_5OH(l)$ are combusted in separate calorimeters. Each combustion reaction occurs in the presence of excess $O_2(g)$ and goes to completion according to the chemical equations shown in the table above. Information about the experiments is shown below.

Experiment	Substance Combusted	Molar Mass	Mass of Substance Combusted
1	$C_4H_{10}(g)$	58.12 g/mol	25.0 g
2	$C_2H_5OH(l)$	46.07 g/mol	?

What mass of $C_2H_5OH(l)$ should be used in experiment 2 in order to release the same quantity of heat (q) that is released in the combustion of 25.0 g $C_4H_{10}(g)$ in experiment 1?

(A) 19.8 g

(B) 25.0 g

(C) 41.7 g

(D) 52.6 g



The structural formula of acetylene, C_2H_2, is shown above. Acetylene gas reacts with $H_2(g)$ to form ethane gas, $C_2H_6(g)$, as represented by the following equation.

$$C_2H_2(g) + 2\ H_2(g) \rightarrow C_2H_6(g) \quad \Delta H° = ?$$

Bond	Bond Energy (kJ/mol)
H–H	432
C–C	347
C≡C	839
C–H	413

6. Based on the table of bond energies shown above, the value of $\Delta H°$ for the reaction $C_2H_2 + 2\ H_2 \rightarrow C_2H_6$ is closest to which of the following?

   (A) $-728\ \text{kJ/mol}_{rxn}$

   (B) $-296\ \text{kJ/mol}_{rxn}$

   (C) $+296\ \text{kJ/mol}_{rxn}$

   (D) $+728\ \text{kJ/mol}_{rxn}$

$$CH_3OH(g) \rightarrow CO(g) + 2\ H_2(g) \quad \Delta H° > 0$$

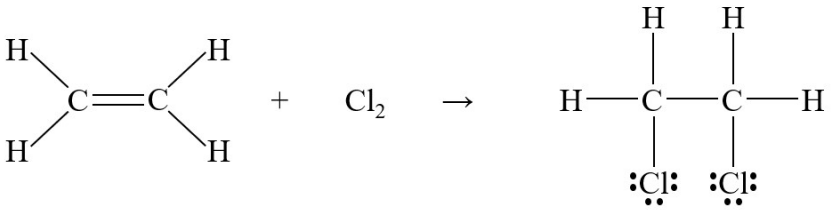
7. Which of the following best explains why the sign of $\Delta H°$ for the reaction represented above is positive?

   (A) The energy absorbed as the bonds in the reactant are broken is greater than the energy released as the bonds in the products are formed.

   (B) The energy released as the bonds in the reactant are broken is greater than the energy absorbed as the bonds in the products are formed.

   (C) The energy absorbed as the bonds in the reactant are broken is less than the energy released as the bonds in the products are formed.

   (D) The energy released as the bonds in the reactant are broken is less than the energy absorbed as the bonds in the products are formed.



ethylene, C_2H_4 1,2-dichloroethane, $C_2H_4Cl_2$

Ethylene, C_2H_4, reacts with chlorine, Cl_2, to produce 1,2-dichloroethane, $C_2H_4Cl_2$, as represented by the equation shown above.

Bond	Bond Enthalpy (kJ/mol)
C–C	347
C=C	614
Cl–Cl	243
C–H	413
C–Cl	339

A student performed the following calculation, using the bond enthalpy values in the table above, to estimate the value of $\Delta H°$ for the formation of 1,2-dichloroethane.

$$\Delta H° = [\ 347 + (4 \times 413) + (2 \times 339)\ ] - [\ 614 + (4 \times 413) + 243\ ] = +168\ \text{kJ/mol}_{rxn}$$

8. What error did the student make in the set-up for this calculation?

(A) The student counted the number of C–H bonds incorrectly.

(B) The bond energy of the chlorine-chlorine bond in Cl_2 should be equal to $(243 \times 2) = 486$

(C) The bond energy for the carbon-carbon bond in C_2H_4 should be equal to $(347 \times 2) = 694$.

(D) The student assumed that energy is released when bonds are broken.

Substance	Standard Enthalpy of Formation (kJ/mol)
$O_2(g)$	0
$SO_2(g)$	–296.8
$SO_3(g)$	–395.7

9. Based on the information in the table above, which of the following choices is closest to the value of $\Delta H°$ for the reaction represented by the following equation?

$$2\ SO_2(g)\ +\ O_2(g)\ \rightarrow\ 2\ SO_3(g) \qquad \Delta H° = ?$$

(A) –197.8 kJ/mol$_{rxn}$

(B) –98.9 kJ/mol$_{rxn}$

(C) +98.9 kJ/mol$_{rxn}$

(D) +197.8 kJ/mol$_{rxn}$

$$2\ CH_3CHO(g)\ +\ 5\ O_2(g)\ \rightarrow\ 4\ CO_2(g)\ +\ 4\ H_2O(l) \qquad \Delta H° = –2390\ \text{kJ/mol}_{rxn}$$

10. The combustion of acetaldehyde, $CH_3CHO(g)$ occurs according to the equation above. Based on the information in the table below and the value of the standard enthalpy change for the combustion of $CH_3CHO(g)$, which of the following is closest to the value of the standard enthalpy of formation for $CH_3CHO(g)$?

Substance	Standard Enthalpy of Formation (kJ/mol)
$CH_3CHO(g)$	?
$O_2(g)$	0
$CO_2(g)$	–394
$H_2O(l)$	–286

(A) –330 kJ/mol

(B) –165 kJ/mol

(C) +165 kJ/mol

(D) +330 kJ/mol

Chemical Equation	$\Delta H°$ (kJ/mol$_{rxn}$)
$N_2(g) + 2\ H_2(g) \rightarrow N_2H_4(l)$	+51
$H_2(g) + \frac{1}{2}\ O_2(g) \rightarrow H_2O(l)$	–286
$2\ H_2O_2(l) \rightarrow 2\ H_2O(l) + O_2(g)$	–196

11. Based on the information in the table above, what is the value of $\Delta H°$ for the following reaction?

$$2\ H_2O_2(l) + N_2H_4(l) \rightarrow 4\ H_2O(l) + N_2(g) \quad \Delta H° = ?$$

(A) –819 kJ/mol$_{rxn}$

(B) –717 kJ/mol$_{rxn}$

(C) –533 kJ/mol$_{rxn}$

(D) –431 kJ/mol$_{rxn}$